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ORIGINAL ARTICLE

Anabolic myokine responses and muscular performance following 8 weeks of autoregulated compared to linear resistance exercise in recreationally active males

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Abstract

Background To date, no studies, to our knowledge, have compared the efficacy of autoregulated periodized and linear resistance exercises on anabolic myokines and muscular performance among recreationally active individuals. This study aimed to compare the effects of an 8-week autoregulated periodized resistance exercise (APRE) program with a linear resistance exercise (LRE) program on insulin-like growth factor-1 (IGF-1), follistatin (FST), myostatin (MST), body composition, muscular strength, and power in recreationally active males.

Methods Thirty males were randomly assigned to either the APRE group ($n = 15$) or the LRE group ($n = 15$). Participants completed training three times a week for 8 weeks. The outcome measures included serum IGF-1, FST, MST, muscular strength (isometric knee extension and handgrip), power (vertical jump), lean body mass, and fat mass.

Results IGF-1 circulating levels increased over time following APRE (34%) and with no significant change following LRE (~-1%). There were no significant differences over time or between groups for FST or MST. Muscular strength (knee extension [21.5 vs. ~16%] and handgrip [right: 31 vs. 25%; left: 31.7 vs. 28.8%]) and power (~33 vs. ~26%) significantly increased to a greater extent following APRE compared to LRE. Interestingly, the results revealed that lean body mass increased over time only after APRE (~3%), but not LRE.

Conclusion These findings suggest that APRE may be more effective than LRE in increasing muscular strength, power, and lean body mass, as well as circulating IGF-1 levels, in recreationally active males. The observed differences may be attributed to the increased training volume associated with APRE. However, further research is needed to directly assess muscle protein synthesis.

Keywords Autoregulation periodization · Follistatin · Insulin-like growth factor-1 · Myostatin

Introduction

Resistance training significantly elevates muscle protein synthesis which, over time, leads to muscle accretion and gains in muscular strength and power [1]. The periodization of resistance training, involving the strategic manipulation of variables such as intensity, frequency, sets, repetitions, velocity, and recovery time, plays a critical role in shaping physiological adaptations [2]. This dynamic adaptation process requires a good understanding of the individual's responsiveness to resistance training, which naturally calls for a more personalized approach [3].

Although linear and non-linear models are two common forms of periodization in resistance exercise [4, 5]. Linear programming has been shown to improve muscular strength and athletic performance [4], some studies suggest that the non-linear model may be more effective in comparison to the linear model for enhancing muscular adaptations induced by resistance exercise [6]. Regarding this issue, Rhea et al. (2002) carried out a study using a non-linear resistance exercise (non-LRE) method called daily undulating programming of resistance exercise. Their findings indicated that this method offers potential advantages over

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LRE in terms of enhancing muscle strength and mitigating training plateaus [7]. Daily undulating programming of resistance exercise comprises variations of intensities from day to day and weekly and seems to be beneficial in that it changes neuromuscular adaptations, such as improvements in muscular fiber recruitment patterns and motor unit synchronization, thus improving recovery [8]. Nevertheless, the majority of studies indicate that both non-LRE and LRE are successful approaches for enhancing muscular strength, even though to date, direct comparisons have yielded uncertain results.

Recently, a new type of exercise called autoregulatory periodized resistance exercise (APRE) has become popular. This approach is an adapted version of the daily-APRE method. One of the primary attributes of APRE is its ability to enhance athletes' performance during training sessions when they have fully recovered physiologically [9]. Specifically, APRE utilizes individualized adjustments, potentially addressing daily fluctuations in performance [10]. In theory, long training periods of structured or fixed blocks of training based on a singular baseline assessment, although individualized, may result in periods of suboptimal loading due to daily fluctuations in an individual's performance. There is a growing body of evidence demonstrating that autoregulated training may augment muscle accretion and muscular strength [10] compared with structured and fixed strategies. In this context, a study investigating two distinct approaches to APRE and LRE during a span of 2 consecutive years reported that APRE showed notable superiority in enhancing both upper and lower body muscular strength [9]. Moreover, following 9 weeks of flexible training in highly trained individuals, Rauch et al. (2020) observed an increase in total training volume, lean body mass, and muscular strength compared to fixed exercise performed by three different mesocycles (day 1: 6–8 RM, day 2: 12–14 RM and day 3: 18–20 RM) [10]. Collectively, autoregulated training appears to benefit whole-body muscle adaptations.

It has been noted that muscle protein balance may be affected by the endocrine system. Various hormones and myokines alter the balance between anabolic and catabolic stimuli in muscle [11]. Follistatin (FST), myostatin (MST), and insulin-like growth factor-1 (IGF-1) appear to impact the upstream pathways of the mechanistic target of rapamycin (mTOR) [12–14]. FST is a potent antagonist that binds directly to MST [15]. IGF-1 binds to receptors on the cell membrane and triggers protein kinase-B (PKB), suppressing forkhead box O (FOXO) transcription factors [16]. IGF-1 also binds to tyrosine kinase receptors, activating and phosphorylating the phosphoinositide 3-kinase (PI3K) and Akt (PKB) signaling pathways. Several studies have investigated the effects of resistance training on IGF-1 with equivocal findings, reporting either an increase [17], a decrease [18], or no change [19, 20]. FST is a monomeric autocrine glycoprotein that may contribute to upregulating muscle hypertrophy [21]. MST is a member of the TGF-β and has a potential negative regulatory role in skeletal muscle growth in animals and humans [13, 22, 23]. Various studies have been conducted on the effects of mentioned markers on mediating muscle hypertrophy following resistance training [24]. Overall, resistance training led to increased FST and decreased MST in trained and inactive populations [25]. In contrast, Willoughby et al. (2004) found that heavy resistance training was associated with increased MST [26]. Furthermore, a substantial increase in MST and a non-significant increase in IGF-1 have been reported after resistance training at 75–80% 1RM accompanied by a high protein diet [27]. Despite widespread support for the positive impact of resistance training on myokine changes, the efficacy of autoregulatory exercise as regards myokine responses remains to be elucidated.

The purpose of this study was to compare the impact of an 8-week APRE with LRE programs on IGF-1, FST, MST, body composition, muscular strength, and power in recreationally active males. Based on literature demonstrating whole body gains in muscle accretion and strength, we hypothesized that APRE would elicit greater alterations in IGF-1, FST, MST, improved body composition, and muscular strength compared with LRE.

Materials and methods

Experimental approach

Anthropometrics and health history were evaluated before the study in order to adequately examine the exercise technique. Following randomization, participants were assigned to two groups. Measurements were collected at two timepoints, namely, at baseline and 8 weeks post-resistance training (48 h after the last exercise bout). Participants had not taken any dietary supplements or medications for at least 6 months prior to enrollment in the study. The resistance training protocols comprised an 8-wk (24 sessions) program.

The participants were, as mentioned, randomized into two groups, namely, APRE ($n = 15$) or LRE ($n = 15$), and the study protocols, risks, and benefits were fully explained to them prior to their signing of informed consent. Body composition, muscular strength and power, and blood samples were measured at baseline and after 8 weeks of either APRE or LRE. The participants were in addition asked to maintain their habitual diet and regular lifestyle throughout the study. Figure 1 presents the schematic design of the study. It is to be noted that the experimental design of this study and

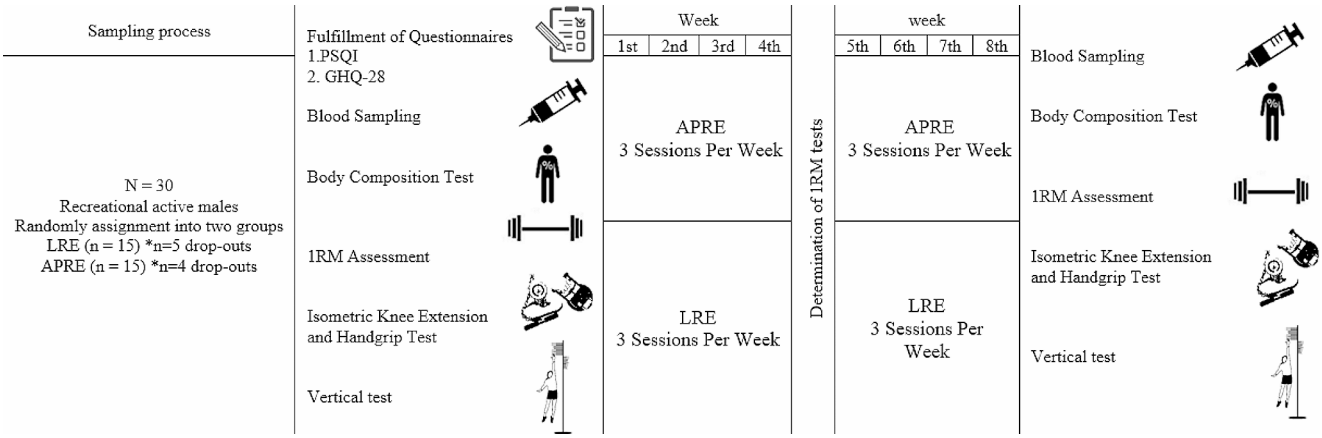

Fig. 1 Schematic of the study design. **Abbreviations**: APRE: Auto-regulatory progressive resistance exercise, LRE, linear resistance exercise; N, sample size; PSQI, Pittsburgh Sleep Quality Index; GHQ-28, General Health Questionnaire-28; 1RM, 1 repetition maximum

a small portion of the methods have been previously published [28], but no duplication exists in any of the dependent variables reported in this study.

Participants

Based on the inclusion criteria, 30 recreationally active males (age: 24.3 ± 3.4 years; weight: 79.2 ± 10.9 kg; body fat: 17.2 ± 2.5%) volunteered for the study. Inclusion criteria were the following: being aged between 20 and 30 and prior resistance training experience (> 6 months). Participants were excluded if they had a sleep disorder, a history of anabolic steroid use, and/or used supplements known to alter muscle biology, and also if participants sustained a musculoskeletal injury within 6 months of the beginning of the study. Participants were additionally removed from the study if they were engaging in other exercise activities outside of the study. Participants completed the Pittsburgh Sleep Quality Index (PSQI) and the General Health Questionnaire-28 (GHQ-28) for assessment of sleep and general health.

Resistance training procedures

Training phase.

After the initial measurements and randomization, participants began training for 8 weeks. Each week, participants performed three training sessions with 48-hour between each. The average duration of training in each session was 62 ± 3.9 (51–78) minutes recorded by the participants. In each session, before performing the main exercises, participants completed a 10-min general warm-up (slow running, stretching, and light resistance training) and a specific warm-up that included two sets of 20 repetitions at 30% of 1-RM with 30 s of rest between sets [29]. In addition, 1-RM (back squat, bench press, lateral pull-down, shoulder press, standing barbell curl, leg curl machine, triceps push-down, and leg extension machine) was determined before and after the 12th and 24th sessions.

Table 1 Autoregulatory progressive exercise protocol

Set	Intensity	Repetitions	Rest (seconds)
Set 1	50% of 1RM	10	60
Set 2	75% of 1RM	6	60
Set 3	85% of 1RM	To failure	60
Set 4	Based on the repetitions completed in 3th set	Adjusted for next session	-

APRE adjustment protocol for fourth sets

Repetitions of third set	Repetitions of fourth set
0–2	-10 lbs (4.5 kg)
3–4	-5 lbs (2.26 kg)
5–7	No change
8–12	+ 5 lbs (2.26 kg)
13+	+ 10 lbs (4.5 kg)

1RM = one repetition maximal. APRE = autoregulatory progressive exercise. Ibs = pounds. Kg = kilogram

APRE protocol

Based on previous research, a 6-RM autoregulation method was selected. This method has been shown to augment muscle accretion and strength gains. Briefly, following a warm-up set, participants completed four more sets, as follows: set 1: 10 repetitions at 50% 6-RM; set 2: six repetitions at 75% 6-RM,; set 3: performed with an intensity of 6-RM until reaching failure; and set 4: adjusted based on the number of repetitions of set 3. The workload did not change if participants completed five to seven repetitions in set 3. However, the workload increased or decreased if the number of repetitions was ≥ 8 or ≤ 4. We subsequently used the number of repetitions completed in set 4 to determine the next session load (Table 1) [30].

LRE protocol

Similarly to APRE, the resistance-training program was completed in four sets. In the first week of the intervention, it started with an intensity of 70% 1RM and increased to a maximum of 85% 1RM, in which the workload increased by 5% consecutively every week. On the other hand, each exercise was performed with three sets of eight repetitions at 70% 1RM during the first and fifth weeks. The training program was implemented during the second and sixth weeks with four sets of six repetitions at 75% 1RM. The number of sets and repetitions was unchanged in the third, fourth, seventh, and eighth weeks and the workload reached 80% and 85% 1RM [9]. The LRE program is outlined in Table 2.

In each training session, training volume was determined by multiplication repetitions, sets, and load. The training volume for each participant was calculated according to the following formula [31]. Training volume = [repetitions × sets × load (kg)]. Training volume data are presented in Fig. 2.

Moreover, in order to measure internal load, the 10-point Borg Ratings of Perceived Exertion (RPE) scale was applied [32]. RPE was recorded 30 min after each training session [33]. RPE data are shown in Fig. 3.

Table 2 Linear resistance exercise (LRE) protocol

Week	Set	Repetitions	Intensity (%)
1RM assessment			
1	3	8	70
2	4	6	75
3	4	5	80
4	4	5	85
1RM assessment			
5	3	8	70
6	4	6	75
7	4	5	80
8	4	5	85
1RM assessment			

LRE = linear resistance exercise. 1RM = one repetition maximum

Measurements

Anthropometry and body composition

We used an Inbody 720 Body Composition Analyzer (South Korea) to measure body composition (body mass, body mass index (BMI), fat mass, and lean body mass). To increase accuracy, the participants were asked to fast for 12 h with at least 8 h of sleep and to refrain from lifting weights prior to the assessment. Upon entering the laboratory, each participant sat on a chair for 30 min, after which their height

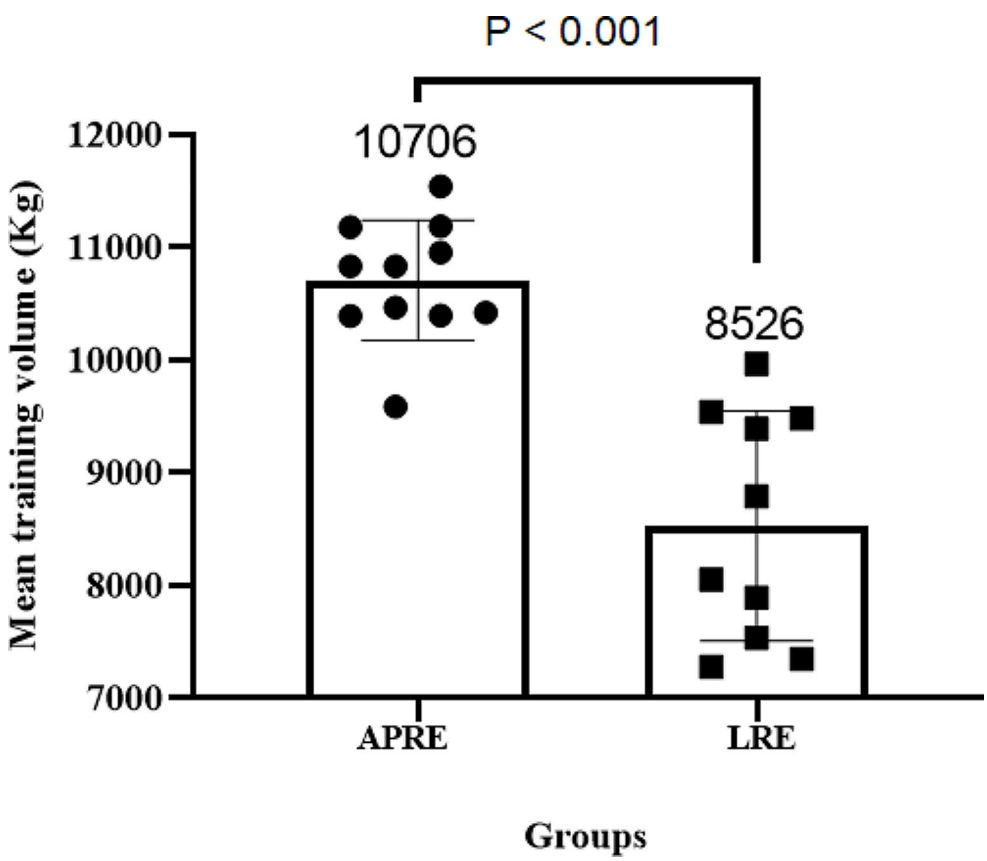

Fig. 2 Mean ± SD of training volume per week in both groups Significant differences between groups based on One-way analysis of variance (ANOVA) test

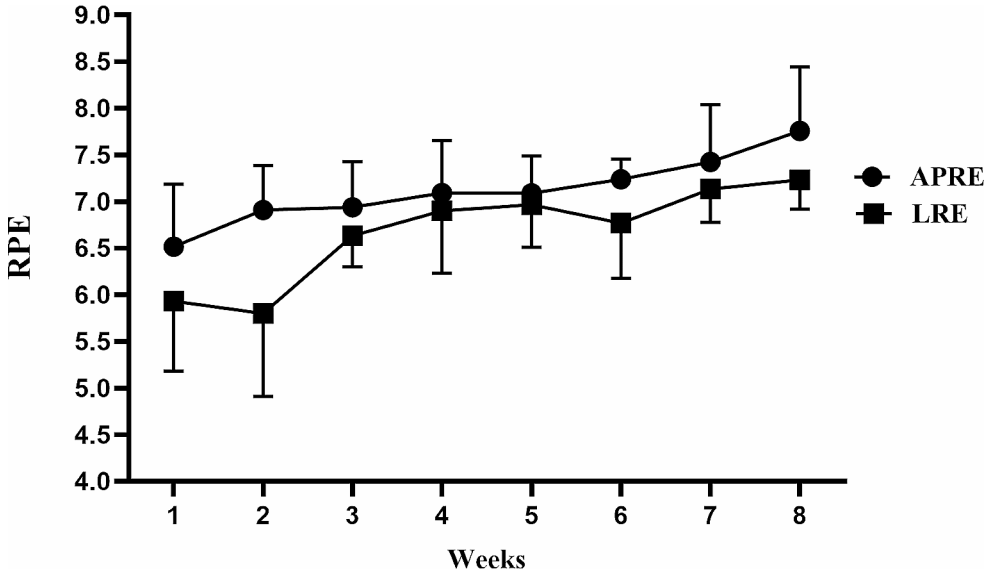

Fig. 3 Mean ± SD of training volume per week in both groups

was measured using a digital stadiometer (Seca, Hamburg, Germany) to the nearest 0.01 m.

Isometric knee extension

Participants conducted an isometric knee extension test with a 5-min rest period in between attempts. The maximum strength of isometric knee extension was evaluated when participants stood on the dynamometer (Takei, Japan) with their knees bent between 130 and 140 degrees and the trunk straight. Participants adjusted the chain's length, so that the dynamometer handle was at mid-thigh level, and gripped it with their hands pronated. They attached the belt to the dynamometer handle on both sides and the shoulder. This setup made it possible to measure the quadriceps isometric force directly. They were given three chances, with a 3- to 5-min break between each attempt. The investigator recorded the highest score and submitted it for analysis [34].

Isometric handgrip

We used a handheld dynamometer to determine isometric handgrip strength in each hand (Yagami, Nagoya, Japan). The elbow of the testing arm was held tightly against the side of the body and bent at 90 degrees. While the handle rested on the middle phalange of digits 2 to 5, we adjusted the dynamometer handle with the base resting on the first metacarpal (heel of the palm). The dynamometer was squeezed by the participants at their maximum isometric effort for a maximum of 5 s. The participants had three chances, with a 3- to 5-min break between each attempt, verbal encouragement meanwhile being given to them. The highest score for each hand was recorded and added together for a total score.

Muscular power - vertical jump test

We applied the vertical jump test to evaluate muscular power. Before initiation of the test, its correct execution was explained to each individual by an expert, after which each participant was requested to stand next to the wall and reach up with the hand closest to the wall. At this moment, the point of the fingertips was marked and used to determine standing reach height. Each participant was then instructed to jump as high as possible. The score was the difference in distance between the standing reach height and the jump height. The best of three attempts was recorded [5]. Height (m) = $(g[9.81\ m/s^2] \times time^2) \div 8$.

Myokines and IGF-1 procedures

Using ZellBio human commercial kits made in Germany, serum IGF-1, FST, and MST were measured with the ELISA (enzyme-linked immunosorbent assay) sandwich method in accordance with the manufacturer's instructions. Blood samples were measured 48 h before and after the training session. After taking blood samples (10 mL) from the cubital vein, these were allowed to clot for 10–15 min at room temperature. Next, the samples were centrifuged for 10 min at 3000 RPM. The supernatants were carefully extracted and frozen at -80 °C for later analysis. The assay range and sensitivity were 1.5–48 ng/mL; 0.06 ng/mL for IGF-1; 7.5–240 ng/mL; 0.25 ng/mL for FST and 75-2400 ng/L; and 0.5 ng/L for MST. The coefficient of variations [35] was calculated by dividing the standard deviation of a set of measurements by the mean for all kits (intra-assay CV < 10% and inter-assay CV < 12%). All participants were instructed to refrain from taking medication and performing strenuous activity

Table 3 Energy and macronutrient data (mean ± SD)

	APRE		LRE	
	Pre	Post	Pre	Post
Energy (kcal/day)	2754.1 ± 76.9	2792.5 ± 90.4	2764.7 ± 81.4	2787.6 ± 94.8
Carbohydrate (g/day)	335.4 ± 14.4	335.9 ± 15.2	335.9 ± 9.1	336.2 ± 10.8
Protein (g/day)	115.8 ± 11.9	120.4 ± 8.7	116.1 ± 8.2	119.5 ± 9.0
Fat (g/day)	105.4 ± 8.1	107.4 ± 4.7	106.3 ± 7.5	107.2 ± 5.2

72 h prior to blood sampling. In addition, they were asked to maintain and replicate their diet 48 h prior to testing.

Dietary analysis

In order to analyze total energy expenditure and the amount of energy derived from macronutrients, the participants were asked to record their diet, which was assessed utilizing Diet Analysis Plus version 10 software (Cengage, Boston, MA, USA). In addition, all participants were reminded not to change their daily diet throughout the investigation.

Statistical analysis

The sample size was determined a priori using G-power software. Briefly, an effect size of 0.5, an alpha level of 0.05 at 80% power, and a correlation of 0.8 between pre- and post-values determined that 24 participants in total (12 per group) were required. Due to potential drop-out, 20% more participants were initially recruited. Prior to data analysis, the normality of the data was judged using the Shapiro-Wilk test. Data are presented as means ± standard deviations. One-way analysis of variance (ANOVA) was applied to calculate differences between groups at baseline for all variables, as well as for training volume. A 2 (groups: APRE) x 2 (time: pre vs. post) repeated measures ANOVA was used to assess changes between groups over time for IGF-1, FST, MST, LBM, fat mass, muscular strength, and power. The Tukey post hoc test was applied to measure differences between group means. Effect sizes were determined based on the partial eta-squared (η^2). Statistical significance was set a priori at $P < 0.05$, although p-values approaching significance (i.e., $0.050 < P < 0.01$) were also discussed given the exploratory nature of the study [36]. All analyses were calculated using SPSS software (version 24) [37].

Results

Dropouts and dietary analysis and training volume

Nine participants were excluded due to either fear of COVID-19 infection or personal issues, resulting in a total of 21 participants completing the study ($n = 11$ APRE; $n = 10$ LRE). Compliance with RT was 70% in both groups. There were no significant differences over time or between groups in mean daily energy intake as well as in the amount of protein, carbohydrates, and fats consumed per day, as shown in Table 3. In addition, training volume was significantly greater ($P < 0.001$) following APRE (10,706 ± 4259 kg) compared to LRE (8526 ± 1071.3 kg).

Table 4 Baseline characteristics of participants

Variables	Groups	Mean ± SD	*P*-value
Age (years)	APRE	24.7 ± 2.7	0.874
	LRE	23.9 ± 4.0	
PSQI	APRE	3.0 ± 1.1	0.925
	LRE	2.8 ± 1.5	
GHQ-28q	APRE	42.6 ± 2.5	0.745
	LRE	44.9 ± 4.3	
Height (cm)	APRE	177.6 ± 7.7	0.865
	LRE	178.2 ± 7.0	
Experience (month)	APRE	8.2 ± 1.6	0.689
	LRE	9.0 ± 2.1	

PSQI = Pittsburgh *Sleep Quality Index. GHQ-28q = The General Health Questionnaire-28. APRE* = autoregulatory progressive exercise. LRE = linear resistance exercise

Participant and anthropometric characteristics

Table 4 depicts participant anthropometric characteristics, including age and height, as well as sleep quality, general health report, and history of training. There was a significant interaction effect for BMI ($P = 0.006$; $\eta^2 = 0.344$), with the post hoc displaying both groups increasing over time ($P \leq 0.001$) to a similar value ($P = 0.99$). Likewise, there was a significant interaction effect for body mass ($P = 0.009$; $\eta^2 = 0.305$), with the post hoc displaying both groups increasing over time ($P < 0.001$) to a similar level ($P = 0.99$). There was a trend towards an interaction effect for lean body mass ($P = 0.084$; $\eta^2 = 0.149$), with forced exploratory post hoc analysis demonstrating that APRE significantly increased over time ($P = 0.002$), while there was no change following LRE ($P = 0.43$). There were no significant interactions or main effects ($P > 0.05$) for fat mass, intracellular water, extracellular water, and total body water (Table 5).

Table 5 Body composition, upper and lower isometric, vertical jump, blood samples and 1RM exercise characteristics in each group at baseline and post-test

Variable	APRE		LRE	
	Pre	Post	Pre	Post
Body composition				
	78.9 ± 11.5	80.7 ± 11.6 [a]	79.4 ± 10.3	80.2 ± 10.7 [a]
LBM (kg)	65.3 ± 9.0	67.2 ± 8.9 [a,b]	65.6 ± 7.0	66.3 ± 7.4 [a]
FM (kg)	13.5 ± 3.5	13.4 ± 4.3	13.7 ± 3.9	13.8 ± 3.9
BMI(kg/m^2)	24.9 ± 2.5	25.2 ± 2.4 [a]	24.9 ± 2.4	25.2 ± 2.4 [a]
ICW(L)	29.8 ± 4.1	30.0 ± 3.8	30.3 ± 3.5	29.3 ± 3.6
ECW(L)	17.6 ± 2.5	17.7 ± 2.5	17.9 ± 2.1	18.0 ± 2.1
TBW(L)		47.6 ± 6.4	48.2 ± 5.6	48.4 ± 5.6
Upper and lower isometric test	47.4 ± 6.6			
Isometric knee extension (N)	92.0 ± 5.3	111.8 ± 4.7 [a,b]	88.7 ± 6.8	103.1 ± 6.4 [a]
Right handgrip (kg)	42.6 ± 2.6	55.9 ± 3.1 [a,b]	40.7 ± 4.2	51.0 ± 4.0 [a]
Left handgrip (kg)	41.9 ± 3.5	55.2 ± 3.5 [a,b]	40.6 ± 4.7	52.3 ± 5.2 [a]
Muscular power				
Vertical jump (cm)	38.0 ± 3.0	50.4 ± 3.4 [a,b]	36.1 ± 4.2	45.6 ± 4.0 [a]
Blood samples				
IGF-1(ng/ml)	14.6 ± 3.6	19.6 ± 6.0 [a,b]	13.0 ± 2.4	12.9 ± 2.6 [a]
FST (ng/ml)	34.6 ± 6.9	40.8 ± 13.0	34.8 ± 8.7	33.9 ± 7.1
MST (ng/ml)	366.6 ± 52.1	374.5 ± 80.2	321.3 ± 58.13	347.4 ± 64.6

BM, body mass; LBM, lean body mass; FM, fat mass; BMI, body mass index; ICW, intra cellular water; ECW, extra cellular water; TBW, total body water; IGF-1, insulin growth factor one; FST, follistatin and MST, myostatin. [a] significantly different than baseline. [b] Post APRE significant difference than post LRE ($p < 0.05$)

Isometric knee extension

A significant interaction was observed for isometric knee extension strength ($P = 0.008$; $\eta^2 = 0.315$). Post hoc analysis revealed that both groups increased over time ($P < 0.001$); importantly, following APRE, isometric knee extension strength was higher than following LRE ($P = 0.021$), as shown in Table 5.

Isometric handgrip

There was an interaction effect for combined grip strength ($P < 0.001$; $\eta^2 = 0.7996$). Post hoc analysis revealed that both groups increased over time ($P < 0.001$); importantly, following APRE, handgrip strength was higher than following LRE ($P = 0.03$), as shown in Table 5.

Muscular power

Vertical jump showed a significant interaction effect ($P = 0.001$; $\eta^2 = 0.795$). Post hoc revealed that both groups significantly increased over time ($P < 0.001$); however, the vertical jump was significantly higher following APRE compared to LRE ($P = 0.03$), as shown in Table 5.

IGF-1, FST, and MST levels

There was a significant interaction for IGF-1 ($P = 0.015$; $\eta^2 = 0.272$). Post hoc analysis revealed a significant increase over time following APRE ($P = 0.006$), while there was no change over time following LRE ($P = 0.99$). There was no significant interaction or main effects ($P > 0.05$) for FST or MST, as shown in Table 5.

Discussion

The aim of the current study was to compare the effects of 8 weeks of APRE and LRE on IGF-1, FST, MST, body composition, muscular strength, and power among recreationally active males. The main findings were that IGF-1 increased over time following only APRE; however, no differences were observed in serum levels of FST and MST either over time or between groups. Moreover, muscular strength and power increased significantly more after APRE compared to LRE. Through exploratory analyses, there appears to be an increase in lean body mass only after APRE.

The present study indicates an increase in lean body mass following APRE, while there was no change over time following LRE. These findings support previous research. For example, Rauch et al. (2020) observed that APRE augmented muscle accretion compared to fixed exercise selection training [10]. Furthermore, Stone et al. (2018) found that 12 weeks of APRE training accompanied by whey protein were associated with increased lean body mass and fat mass [38]. In addition, in the latter study, APRE led to greater gains in muscular strength. These findings support Mann et al. (2010), who reported that APRE training elicited greater muscular strength gains than LRE [9]. Stone et al. (2018) demonstrated that a 12-week APRE program accompanied by whey protein supplementation in untrained persons led to an enhancement in muscular strength [39]. Overall, these

results suggest that adjusting for daily individual performance (i.e., autoregulated training) may expand resistance training adaptations (i.e., lean body mass and muscular strength). It is widely documented in the literature that resistance training leads to an increase in muscle mass in the majority of healthy individuals [40]. Furthermore, greater muscle gain is achieved through higher training volumes [41, 42], completing more sets [43], and specifically targeting the muscle groups involved in the training [44].

Beyond whole body changes, the present study demonstrated a significant elevation in IGF-1 after an 8-week APRE regimen. IGF-1 is a potential regulator of muscle hypertrophy that directly induces phosphorylation, activates the Akt/mTOR pathway [45], and promotes differentiation of proliferating satellite cells. Moreover, IGF-1 binds to its receptor, activating either the Ras-Raf-MEK-ERK or mitogen-activated protein kinase (MAPK) signaling pathways. The Akt/mTOR pathway is responsible for cell differentiation, whereas the Ras- extracellular signal-regulated kinase 1/2 (ERK1/2) pathway seems to be related to cell proliferation [46]. Acute resistance training upregulated factors involved in myogenic activation of satellite cells, IGF-1, IGF-R1 (receptor 1), and mechano growth factors (MGF) [47]. IGF-1 alterations following chronic resistance training programs have previously been investigated yielding mixed findings. In weightlifters, following chronic resistance training, circulating IGF-1 concentrations increased significantly [48]. Borst et al. (2001) demonstrated that 13 weeks of resistance training increased IGF-1 by 20% after both single and multi-set training, but no further increases were found between 13 and 25 weeks [49]. Taken together, these findings clearly point to a number of possible mechanisms whereby resistance training may enhance IGF-1 availability for skeletal muscle and indicating that this may be related to the bioavailability of its binding protein (IGFBP-3) [50]. IGF-1 responses are associated with blood lactate and greater acidity [51], the intensity of training [47], metabolic stress [52], muscle damage [48], muscle contraction [53], and duration of training [54]. In the current study, training volume was higher following APRE, which may have contributed to the greater IGF-1 responses. However, whether IGF-1 is directly associated with gains in muscle mass over time is controversial. For instance, Walker et al. (2004) reported an increased cross-sectional area in elbow flexor muscles without significantly increasing IGF-1 concentrations following a 10-week resistance training program [54]. On the other hand, Mitchell et al. (2013) concluded that variations of circulating hormones such as IGF-1 and testosterone do not play a major role in hypertrophy, which they postulated was primarily controlled by intrinsic mechanisms such as interleukin 6 (IL-6), ribosomal protein S6 kinase beta-1 (S6K1) phosphorylation, and changes in muscle androgen receptor protein content [55]. The present study revealed a significant increase in IGF-1 after APRE compared to LRE, and APRE appeared to increase LBM; nevertheless, the cause and effect of IGF-1 on muscle accretion remain to be fully elucidated.

In this study we found 17.92% increases in serum levels of FST in the APRE group, as well as an increase in MST (2.15% vs. 8.12% for APRE and LRE, respectively), although they were not significant, contrasting with prior research showing increased FST or decreased MST following resistance training. For instance, Farzanegi et al. (2021) reported that resistance training at 70% 1-RM induced increases in FST and decreases in MST in sedentary men [25]. Similarly, Saremi et al. (2010) noted a significant reduction in MST and possibly an increase in G-protein coupled receptor-associated sorting protein 1 (GASP-1), suggesting inhibition of MST through binding to activin IIb receptor following resistance training [56]. Conversly, Willoughby et al. (2004) noted a 56% increase in serum MST levels after 12 weeks of heavy resistance training, which they attributed to eccentric contractions inducing muscle damage, with concurrent observations of changes in fat-free mass and muscle strength [26]. Methodological differences, such as age, fat mass, blood sampling time, and exercise parameters (i.e., duration, intensity, and type) may contribute to these discrepancies. Two potential sources of MST, namely, the local pool of MST in skeletal muscle and the system pool in plasma, have been identified. The local pool is released following sarcomere and sarcolemma damage [57], playing a crucial role in the muscle synthetic protein response to repeated minor muscular damage after eccentric contractions. The mean session ratings of perceived exertion in APRE and LRE groups was 7.13 and 6.73, respectively, highlighting the effectiveness of the APRE versus LRE to increase effort and training volume. The results of the current study thus strongly suggest that the APRE utilized is more efficient than LRE, possibly due to its ability to increase training volume. This increase in volume is often acknowledged as a key factor in promoting hypertrophic adaptations. The latter results differ from those of earlier studies on linear and non-linear programming resistance training, which have indicated equivalent training volume [58–60]. It is important to mention that in our study, participants had a background of resistance training although exclusively engaging in recreational physical activity.

Our study has a number of limitations that should be considered. First, we used bioelectrical impedance to evaluate body composition, which is not as accurate as dual-energy X-ray absorptiometry (the gold standard in this regard); nevertheless, previous research has indicated that it can be considered as a valid and reliable technique [61]. Additionally, recreationally active participants were recruited; thus

the current findings cannot be generalized to either well-trained or untrained persons.

Conclusion

In conclusion, our study demonstrates that 8 weeks of APRE, but not LRE, resulted in an IGF-1 increase over time; however, no differences were observed in serum levels of FST and MST either over time or between groups. Notably, muscular strength and power increased significantly more after APRE compared to LRE. These findings may be associated with increased training volume with APRE compared to LRE. Overall, APRE appears to be more effective than LRE in recreationally active males. Further studies directly assessing muscle protein synthesis are required. Consequently, APRE is the preferred training strategy to increase gains in muscle performance and should be considered by strength and conditioning practitioners.

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Declarations

Ethical approval The study design and protocols were registered (IRCT20191231045963N1) and the ethics were approved by the Iran National Committee for Ethics in Biomedical Research (IR.UM.REC.1398.099) of the Ferdowsi University of Mashhad; the study was conducted in accordance with the ethical standards of the Deceleration of Helsinki.

Informed consent Before participation in the study, written informed consent was obtained from all volunteer participants.

Conflict of interest The authors declare no competing interests.

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